Use work-energy to solve motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose work-energy when force and displacement are known.

SUCCESS CRITERIA - I can...

1. List all forces doing work.
2. Calculate the kinetic-energy change.
3. Check whether the result is physically possible.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “kinetic energy” to explain your model or result. Sentence starters are optional.

work: energy transferred by a force along a displacement

kinetic energy: energy associated with motion

final: at the end of the interval; subscript f

01 Warm-up: bring back an earlier idea

A 4 N force acts forward through 6 m.
Find work and show the force/displacement directions.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $W_{net} = ½m(v_f^2 - v_i^2)$.
Constant mass. Use the displacement along each force component; stopping removes the initial kinetic energy.

Worked reasoning: A 2 kg cart starts from rest. Constant net force 4 N acts through 3 m, doing 12 J. Then $½mv^2=12$ gives $v=\sqrt{12}\approx3.46$ m/s.

Your turn: Sketch a force diagram and an energy bar diagram for this motion.
Identify where time is unnecessary.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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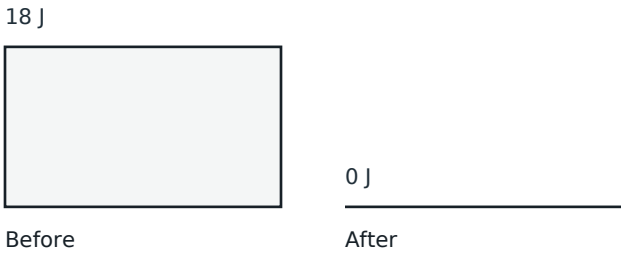
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A cart has 18 J kinetic energy and then stops. How much net work is done?
Show an energy bar before and after.

Sentence starter: I notice … . This supports … because … .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: 1 2 3

04 Apply the physics

The cart in question 3 has 18 J of kinetic energy. A constant 4 N resistive force brings it to rest.
Show the energy removed and calculate the stopping distance.
Explain why no time measurement is needed.

Sentence starter: My claim is … . The diagram/data show …, so … .



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A calculation gives negative final kinetic energy after a long braking distance. What should you check?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Two carts approach: one has momentum 6 kg·m/s east and the other 4 kg·m/s west.
Explain why adding their magnitudes gives the wrong total; find the signed total.
Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, including direction

Sentence starter: The step I would change is … . The evidence is … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A cart has 12 J initial kinetic energy and gains 8 J net work.
Find final kinetic energy.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A cart needs 18 J to stop.
Compare stopping distances for constant opposing forces 3 and 6 N.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3